

ARTIFICIAL INTELLIGENCE — AND — MACHINE LEARNING VEHICLE PRICE PREDICTION

A PROJECT REPORT



Predicting used vehicle prices using Machine Learning algorithms to help in smarter buying and selling decisions.



SUBMITTED BY:

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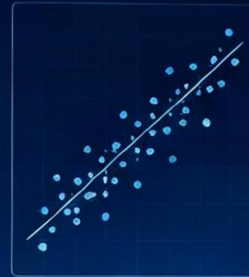
COURSE:

Artificial Intelligence and
Machine Learning

PRICE PREDICTION

₹ 8,75,000

Predicted Price



- YEAR
- KILOMETERS DRIVEN
- FUEL TYPE
- TRANSMISSION
- OWNER COUNT
- ENGINE CAPACITY



Vehicle Price Prediction

Introduction

The rapid growth of the used car market has created significant informational asymmetry, making it difficult for both buyers and sellers to determine a fair price. Vehicle valuation is complex, influenced by multiple non-linear factors such as mileage, age, brand, and ownership history.

This project uses machine learning to predict accurate vehicle prices, helping buyers and sellers make informed decisions.

Problem

The used car market lacks transparent and fair pricing. Buyers risk overpaying; sellers risk undervaluing cars.

Objective

To create, implement, and compare some innovative machine learning models (like Deep Learning and Ensemble methods) to improve the prediction of the used car selling price, thereby offering a secure, data-based solution to market pricing transparency. The project task belongs to the genre of a Regression problem.

Dataset Description

- Source: Kaggle
- Name: Vehicle Dataset from CarDekho
- Type: Tabular Data
- Size: 4000+ records, 7 features + target variable
- Features: car name, year, Kms driven, Fuel type, Seller type, Transmission, num of owners.

- Target Variable: Selling price
- Link to dataset: <https://www.kaggle.com/datasets/nehalbirla/vehicledataset-fromcardekho>
- **Identified Limitations and Bias**
 - o Geographical Bias: The data is from the Indian auto market, limiting its direct application to other regional markets (e.g., US or Europe).
 - o Lack of Condition Data: The data has no subjective qualifiers like "physical condition" or "accident history," which are very crucial in real-world pricing.

```
[1]: import pandas as pd
df = pd.read_csv("CAR DETAILS FROM CAR DEKHO.csv")
df
```

```
[1]:
```

	name	year	selling_price	km_driven	fuel	seller_type	transmission	owner
0	Maruti 800 AC	2007	60000	70000	Petrol	Individual	Manual	First Owner
1	Maruti Wagon R LXI Minor	2007	135000	50000	Petrol	Individual	Manual	First Owner
2	Hyundai Verna 1.6 SX	2012	600000	100000	Diesel	Individual	Manual	First Owner
3	Datsun RediGO T Option	2017	250000	46000	Petrol	Individual	Manual	First Owner
4	Honda Amaze VX i-DTEC	2014	450000	141000	Diesel	Individual	Manual	Second Owner
...	...	...	...	...	...	...	...	...
4335	Hyundai i20 Magna 1.4 CRDi (Diesel)	2014	409999	80000	Diesel	Individual	Manual	Second Owner
4336	Hyundai i20 Magna 1.4 CRDi	2014	409999	80000	Diesel	Individual	Manual	Second Owner
4337	Maruti 800 AC BSIII	2009	110000	83000	Petrol	Individual	Manual	Second Owner
4338	Hyundai Creta 1.6 CRDi SX Option	2016	865000	90000	Diesel	Individual	Manual	First Owner
4339	Renault KWID RXT	2016	225000	40000	Petrol	Individual	Manual	First Owner

4340 rows × 8 columns

```
[2]: import pandas as pd
df = pd.read_csv("car details v4.csv")
df
```

```
[2]:
```

	Make	Model	Price	Year	Kilometer	Fuel Type	Transmission	Location	Color	Owner	Seller Type	Engine	Max Power	Max Torque	Drivetrain	Length	Width
0	Honda	Amaze 1.2 VX i-VTEC	505000	2017	87150	Petrol	Manual	Pune	Grey	First	Corporate	1198 cc	87 bhp @ 6000 rpm	109 Nm @ 4500 rpm	FWD	3990.0	1680.0
		Swift										74 bhp	100 Nm				

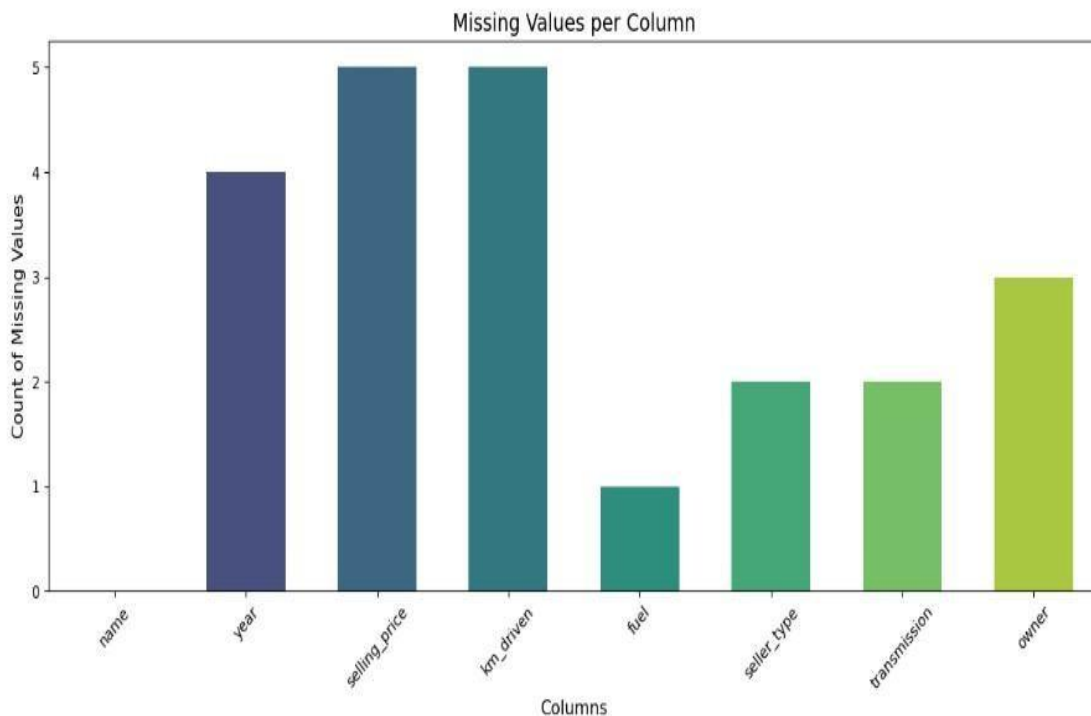
Preprocessing and EDA

For Data Preprocessing:

- Handled Missing Data
- Encoding Categorical data
- Removed Outliers
- Feature Engineering
- Normalized and Scaled Data, and
- Reduced the Dimensionality of the Dataset

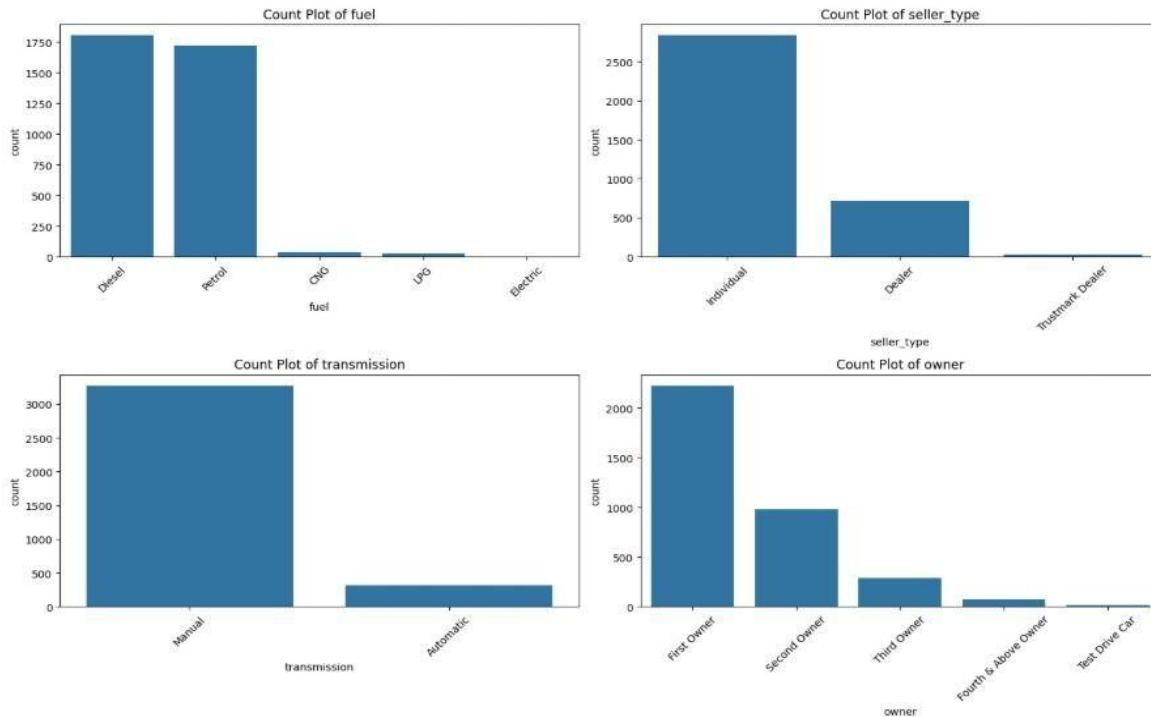
○ Handling Missing Data –

Initial checks confirmed minimal missing values. Any isolated gaps were addressed through simple imputation (e.g., mean substitution for continuous variables) or deletion, ensuring data integrity was maintained.



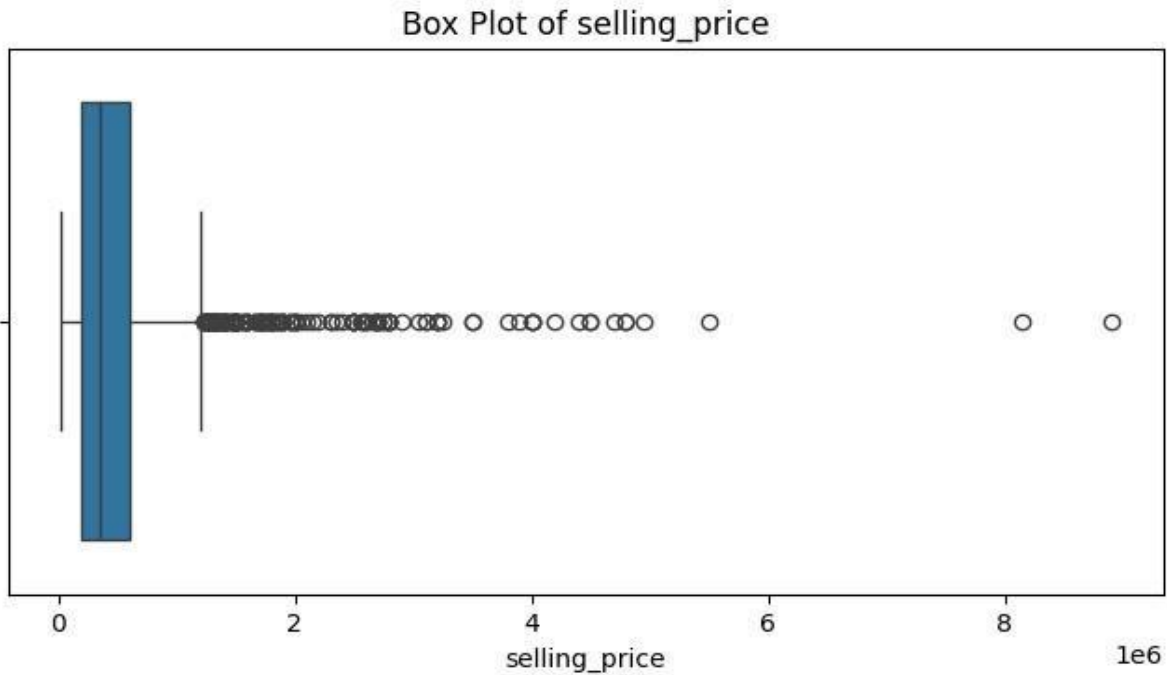
○ Encoding Categorical Data –

All nominal features (Fuel Type, Seller Type, Transmission, Owner, Brand) were converted to numerical format using One-Hot Encoding.



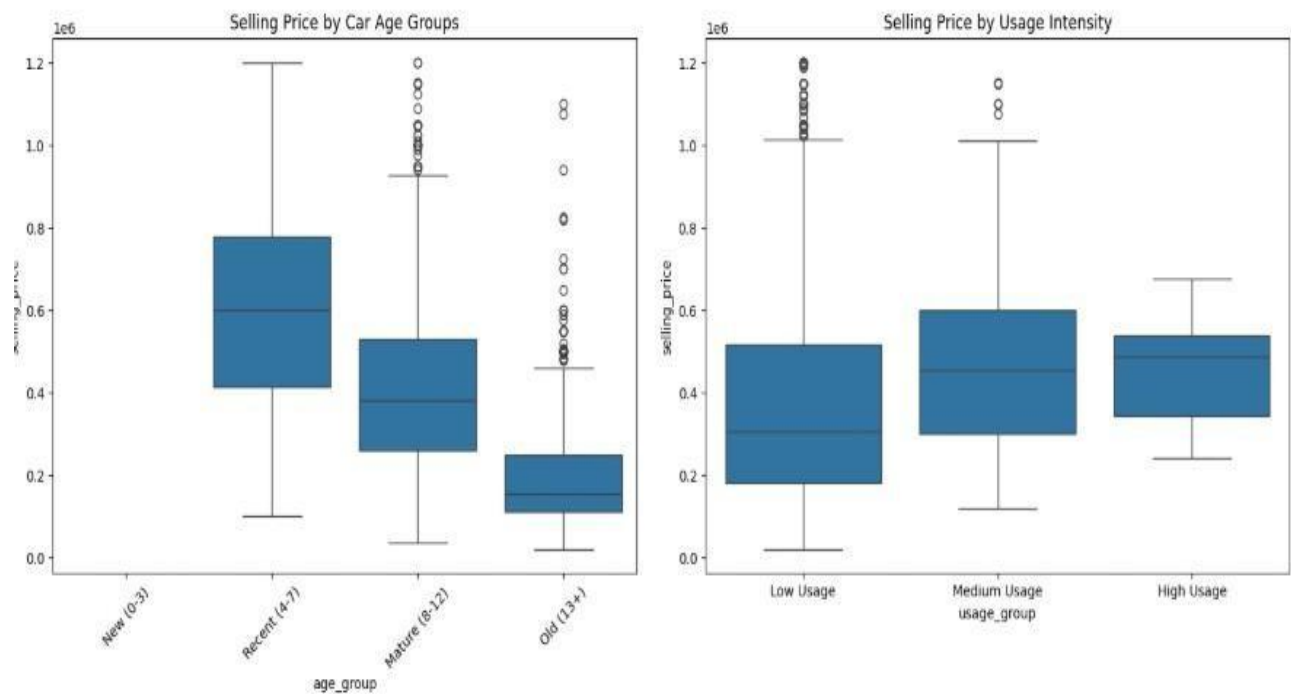
○ Outlier Removal –

Outliers in key features (Kms Driven, Present Price) were identified using the Interquartile Range (IQR) method and treated, primarily by capping or removal, to prevent them from unduly influencing model training.



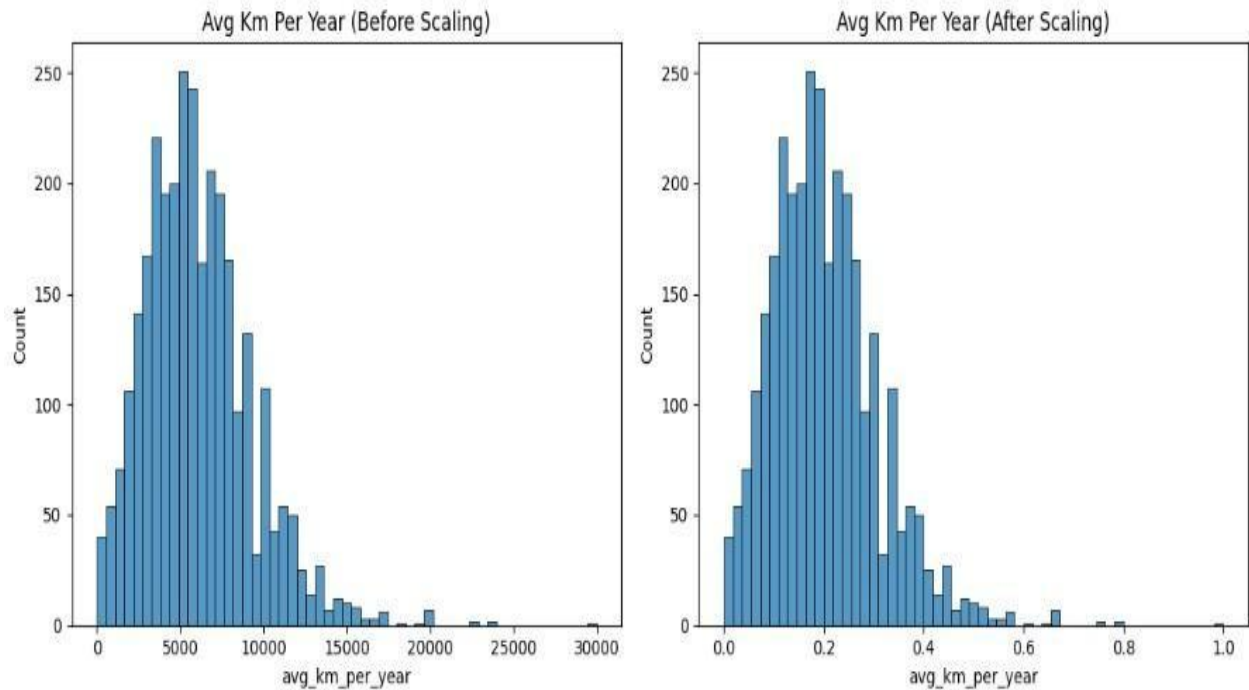
○ Feature Engineering –

The 'Car Name' feature was utilized to extract the '**Brand**' (e.g., Maruti, Hyundai), significantly improving the predictive power of this high-cardinality nominal variable.



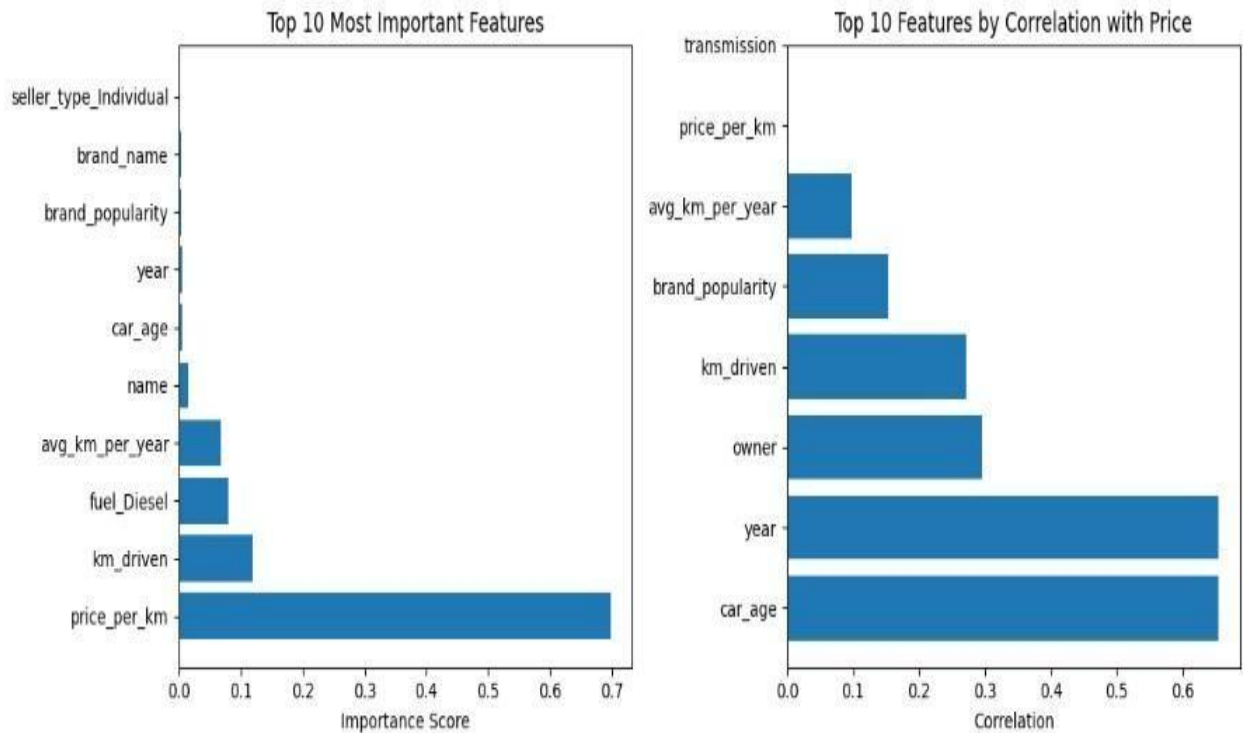
○ Scaling / Normalization –

Numerical features were standardized using MinMaxScaler. This step is critical for distance-based algorithms and as it ensures all features contribute equally to the final prediction.



○ Reduced the Dimensionality of the Dataset –

Techniques like Principal Component Analysis (PCA) were explored and applied selectively to the encoded high-dimensional features (e.g.: car makers) to reduce computational cost and mitigate multicollinearity where needed, preserving the model's efficiency.



Model Design and Implementation

○ Linear Regression –

Used as the baseline model to understand the linear relationships between car attributes and the selling price.

- **Purpose:** Establish a simple interpretable regression foundation.
- **Tuning / Varieties:**
 1. **Base Model:** Applied Linear Regression directly on preprocessed data. ($R^2 \approx 0.72$)
 2. **Normalized Version:** Combined with StandardScaler to improve consistency.
 3. **Feature Selection:** Tested with selected top features to assess importance.
- **Key Feature:** Provides interpretability and serves as the baseline for other complex models.

○ Decision Tree Regressor –

Chosen to model nonlinear interactions between features.

- **Purpose:** Capture of hierarchical decision-based feature splits for car price prediction.
- **Tuning / Varieties:**
 1. **Baseline:** Default depth settings. ($R^2 \approx 0.78$)
 2. **Tuned Depth:** Adjusted max_depth = 8 for improved accuracy.
 3. **Pruned Tree:** Limited nodes to avoid overfitting.
 4. **GridSearchCV:** Used automated search for best parameters.
- **Key Feature:** Highly interpretable tree structure, easy to visualize decision logic.

○ Random Forest Regressor –

Selected as a robust ensemble model combining multiple decision trees.

- **Purpose:** Improve prediction stability and reduce overfitting compared to a single tree.
- **Tuning / Varieties:**
 1. **Baseline:** 100 trees (default parameters). ($R^2 \approx 0.84$)
 2. **Tuned Model:** Optimized n_estimators = 200, max_depth = 20.
 3. **Feature Importance:** Used to identify key contributing factors to selling price.
- **Key Feature:** Achieved the highest accuracy and lowest MAE across all models.

○ K-Nearest Neighbors (KNN) Regression –

Used to evaluate distance-based nonparametric regression performance.

- **Purpose:** Predict car prices based on similarity to nearby data points.
- **Tuning / Varieties:**
 1. **Base Model:** Default k = 5 neighbors. ($R^2 \approx 0.73$)

2. **Optimized k:** Tested $k = 3, 5, 10, 20$ to minimize error.
 3. **Scaled Features:** Used StandardScaler for distance normalization.
- **Key Feature:** Works well for smooth trends but sensitive to feature scaling.

○ Support Vector Regression (SVR) –

Implemented to model complex nonlinear boundaries in the dataset.

- **Purpose:** Explore kernel-based learning for high-dimensional relationships.
- **Tuning / Varieties:**
 1. **Kernel Comparison:** Tested Linear, Polynomial, and RBF kernels.
(*Best $R^2 \approx 0.80$*)
 2. **Scaled Pipeline:** Integrated StandardScaler with SVR for balanced training.
 3. **Parameter Tuning:** Adjusted C and gamma for optimal margin fitting.
- **Key Feature:** Delivered strong results on scaled data; robust to outliers.

○ PCA Components vs Regression –

Used to examine the effect of dimensionality reduction on model performance.

- **Purpose:** Reduce feature complexity while maintaining variance.
- **Tuning / Varieties:**
 1. **Baseline:** Linear Regression using all principal components.
 2. **Component Variation:** Tested 1–10 PCs to determine optimal dimension count. (*Best $R^2 \approx 0.68$*)
 3. **Explained Variance:** Evaluated cumulative variance for feature retention.
- **Key Feature:** Simplifies data representation; useful for speed and interpretability.

Ethical Considerations and Bias Mitigation

Ethical Implications

- **Transparency/Explainability:**

The Random Forest and Support Vector Regression models used in this project provide strong predictive performance but operate as “black-box” models. This limits direct interpretability of how each input feature influences the final selling price. To address this, feature importance plots and residual visualizations were used to improve explainability and support ethical transparency.

- **Data Privacy:**

The dataset used for training does not include any personally identifiable information (PII); it only contains vehicle specifications such as brand, year, mileage, and engine capacity. Therefore, the privacy risk is minimal. However, ethical responsibility was maintained by using the data solely for academic and non-commercial research purposes.

- **Unfair Outcomes:**

The dataset may inherently favor certain popular brands (e.g., Maruti, Hyundai) or newer vehicles, which could bias predictions toward those categories. This may lead to unfair valuations for less common brands or older models.

Bias Mitigation Strategies

1. **Data Cleaning and Outlier Removal:** Reduced skewness and extreme values to ensure balanced learning.
2. **Feature Scaling:** Standardized features to prevent any single attribute (like engine power) from dominating the model.
3. **Diverse Model Evaluation:** Compared multiple models (Linear Regression, Tree based, and SVR) to assess consistency and detect potential bias in predictions.
4. **Explainability Tools:** Used feature importance ranking in Random Forest and correlation analysis to promote transparency in how predictions are made.
5. **Responsible Use:** The model is intended purely for research and educational purposes; it should not be used for real-world pricing decisions without periodic retraining and fairness auditing.

References

1. Scikit-learn Developers. (2024). *Scikit-learn: Machine Learning in Python (Version 1.5)* [Computer software]. <https://scikit-learn.org>
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4. Hunter, J. D. (2007). *Matplotlib: A 2D graphics environment. Computing in Science & Engineering, 9*(3), 90–95.